ASSIGNMENT 6: IMPLEMENTATION OF DATA INTEGRITY CONSTRAINTS

DATABASE MANAGEMENT SYSTEM

ALLIANCE UNIVERSITY

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PROJECT TITLE: Online Food Delivery System

1. Aim

To implement primary keys, foreign keys, UNIQUE, CHECK, and DEFAULT constraints in the Online Food Delivery System and verify that data integrity and relationships between tables are maintained.

2. Software Requirements

- DBMS: MySQL
- Tool: MySQL Workbench
- Language: SQL

3. Theory

Data integrity ensures that the information stored in a database remains accurate, valid, and consistent.

Constraints are rules applied to table columns to prevent invalid data from being inserted or updated.

The major constraints used in this project are:

1. PRIMARY KEY: Uniquely identifies every record in a table and does not allow NULL values.
2. FOREIGN KEY: Ensures that a referenced record exists in the related parent table.
3. UNIQUE: Prevents duplicate values in a column or combination of columns.
4. CHECK: Ensures that data satisfies a specified condition.
5. DEFAULT: Assigns a predefined value when a column is omitted from an INSERT statement.

4. Implementation of Constraints Using SQL

The following SQL statements use the database and tables created in Assignment 5.

1. PRIMARY KEY

Each table already has a primary key.

Input:

```
113 •   SHOW KEYS FROM Customer WHERE Key_name = 'PRIMARY';
114 •   SHOW KEYS FROM Orders WHERE Key_name = 'PRIMARY';
```

Output:

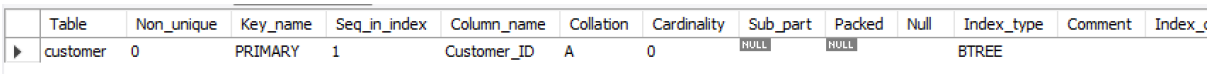


Table	Non_unique	Key_name	Seq_in_index	Column_name	Collation	Cardinality	Sub_part	Packed	Null	Index_type	Comment	Index_c
customer	0	PRIMARY	1	Customer_ID	A	0	NULL	NULL		BTREE		

2. FOREIGN KEY

Verify relationships between tables.

Input:

```
116 •   SHOW CREATE TABLE Address;
117 •   SHOW CREATE TABLE Food_Item;
118 •   SHOW CREATE TABLE Orders;
119 •   SHOW CREATE TABLE Order_Item;
120 •   SHOW CREATE TABLE Payment;
121 •   SHOW CREATE TABLE Delivery;
```

Output:

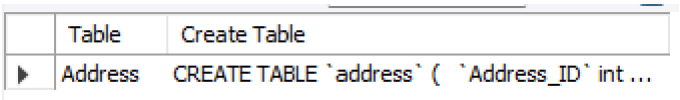


Table	Create Table
Address	CREATE TABLE `address` (`Address_ID` int ...

3. UNIQUE constraints

Customer email must be unique.

Input:

```
123 •   ALTER TABLE Customer
124     ADD CONSTRAINT uq_customer_email UNIQUE (Email);
```

Output:



⚠	10	11:52:01	ALTER TABLE Customer ADD CONSTRAINT uq_customer_email UNIQUE (Email)	0 row(s) affected, 1 warning(s): 1831 Duplicate index 'uq_customer_email' defined on the table 'onlin...	0.062 sec

Category names must be unique.

Input:

```
126 •   ALTER TABLE Category
127     ADD CONSTRAINT uq_category_name UNIQUE (Category_Name);
```

Output:

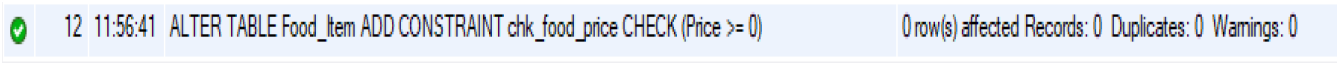


4. CHECK constraints

Input:

```
129 •   ALTER TABLE Food_Item
130     ADD CONSTRAINT chk_food_price CHECK (Price >= 0);
```

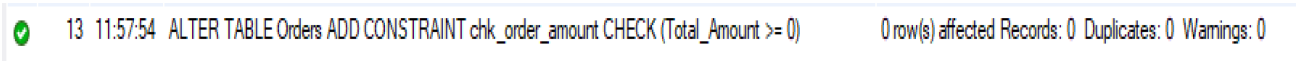
Output:



Input:

```
132 •   ALTER TABLE Orders
133     ADD CONSTRAINT chk_order_amount CHECK (Total_Amount >= 0);
```

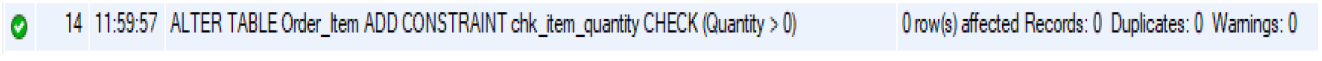
Output:



Input:

```
135 •   ALTER TABLE Order_Item
136     ADD CONSTRAINT chk_item_quantity CHECK (Quantity > 0);
```

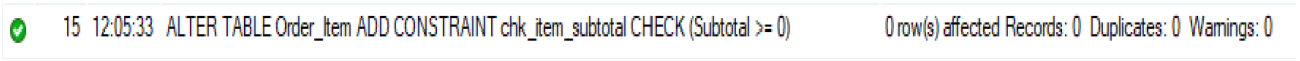
Output:



Input:

```
138 •   ALTER TABLE Order_Item
139     ADD CONSTRAINT chk_item_subtotal CHECK (Subtotal >= 0);
```

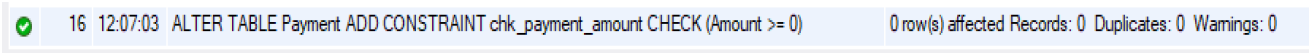
Output:



Input:

```
141 •   ALTER TABLE Payment
142     ADD CONSTRAINT chk_payment_amount CHECK (Amount >= 0);
```

Output:

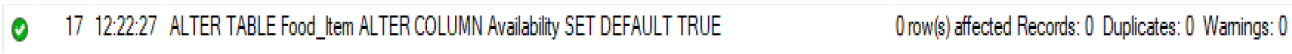


5. DEFAULT constraints

Input:

```
144 •   ALTER TABLE Food_Item
145     ALTER COLUMN Availability SET DEFAULT TRUE;
```

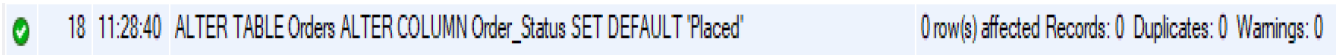
Output:



Input:

```
147 •   ALTER TABLE Orders
148     ALTER COLUMN Order_Status SET DEFAULT 'Placed';
```

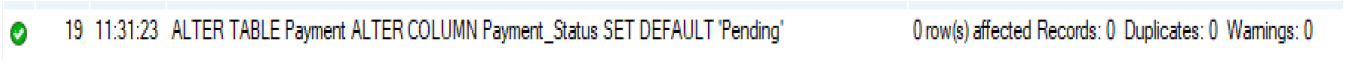
Output:



Input:

```
150 •   ALTER TABLE Payment
151     ALTER COLUMN Payment_Status SET DEFAULT 'Pending';
```

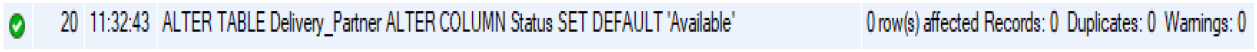
Output:



Input:

```
153 •   ALTER TABLE Delivery_Partner
154     ALTER COLUMN Status SET DEFAULT 'Available';
```

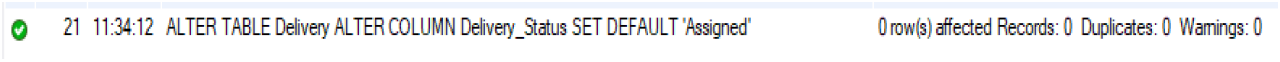
Output:



Input:

```
156 •   ALTER TABLE Delivery
157     ALTER COLUMN Delivery_Status SET DEFAULT 'Assigned';
```

Output:



5. Purpose of Constraints Used

Constraint	Example	Purpose
PRIMARY KEY	Customer_ID	Uniquely identifies each customer
FOREIGN KEY	Orders.Customer_ID	Connects an order to an existing customer
UNIQUE	Customer.Email	Prevents duplicate customer email addresses
CHECK	Quantity > 0	Prevents zero or negative food quantities
CHECK	Price >= 0	Prevents negative food prices
DEFAULT	Order_Status = 'Placed'	Automatically assigns an initial order status
DEFAULT	Payment_Status = 'Pending'	Assigns a default payment status

6. Verification of Data Integrity

The following test statements verify whether the constraints are working correctly.

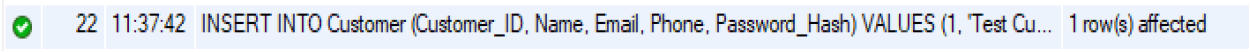
Run each test separately. Some statements are intentionally invalid and are expected to produce errors.

6.1 Test PRIMARY KEY

Input:

```
159 •   INSERT INTO Customer
160     (Customer_ID, Name, Email, Phone, Password_Hash)
161     VALUES
162     (1, 'Test Customer', 'test1@example.com',
163      '9876543210', 'sample_hash');
```

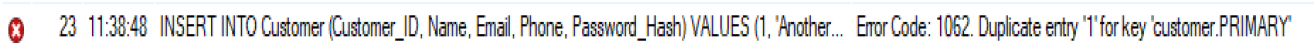
Output:



Input:

```
165 •    INSERT INTO Customer
166     (Customer_ID, Name, Email, Phone, Password_Hash)
167     VALUES
168     (1, 'Another Customer', 'test2@example.com',
169      '9876543211', 'sample_hash');
```

Output:



Expected result: The second insertion fails because primary key values must be unique.

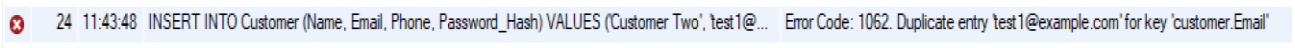
Run this test on a clean database or use an unused ID for the first insertion.

6.2 Test UNIQUE Constraint

Input:

```
171 •   INSERT INTO Customer
172     (Name, Email, Phone, Password_Hash)
173     VALUES
174     ('Customer Two', 'test1@example.com',
175      '9876543212', 'sample_hash');
```

Output:



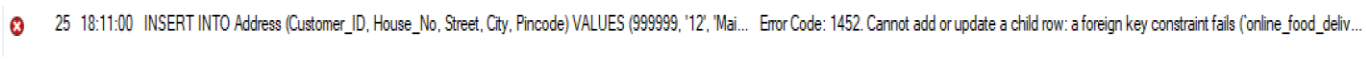
Expected result: The insertion fails because the email already exists.

6.3 Test FOREIGN KEY Constraint

Input:

```
177 •   INSERT INTO Address
178     (Customer_ID, House_No, Street, City, Pincode)
179     VALUES
180     (999999, '12', 'Main Road', 'Chennai', '600001');
```

Output:



Expected result: MySQL rejects the insertion because the referenced customer does not exist.

6.4 Test CHECK Constraint

Input:

```
182 •   INSERT INTO Order_Item
183     (Order_ID, Food_ID, Quantity, Subtotal)
184     VALUES
185     (1, 1, -2, 100.00);
```

Output:

⊗	26	18:14:30	INSERT INTO Order_Item (Order_ID, Food_ID, Quantity, Subtotal) VALUES (1, 1, -2, 100.00)	Error Code: 3819. Check constraint 'chk_item_quantity' is violated.

Expected result: The insertion fails because Quantity must be greater than zero. This test assumes MySQL 8.0.16 or later and existing parent records for Order_ID and Food_ID.

6.5 Test DEFAULT Constraint

Input:

```
187 •   INSERT INTO Orders
188     (Customer_ID, Address_ID, Total_Amount)
189     VALUES
190     (1, 1, 250.00);
```

Output:

⊗	27	18:17:37	INSERT INTO Orders (Customer_ID, Address_ID, Total_Amount) VALUES (1, 1, 250.00)	Error Code: 1452. Cannot add or update a child row: a foreign key constraint fails (`online_food_deliv...

Input:

```
193 •   SELECT Order_ID, Order_Status
194     FROM Orders
195     ORDER BY Order_ID DESC
196     LIMIT 1;
```

Output:

	Order_ID	Order_Status
*	NULL	NULL

Expected result: The newly inserted order receives the default status Placed.

7. Additional Verification Queries

Display customer information

Input:

```
198 •    SELECT * FROM Customer;
```

Output:

	Customer_ID	Name	Email	Phone	Password_Hash
▶	1	Test Customer	test1@example.com	9876543210	sample_hash
*	NULL	NULL	NULL	NULL	NULL

Display order details

Input:

```
200 •    SELECT * FROM Orders;
```

Output:

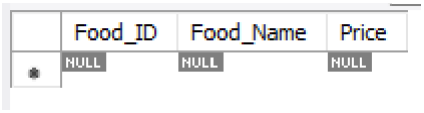
	Order_ID	Customer_ID	Address_ID	Order_Date	Total_Amount	Order_Status
*	NULL	NULL	NULL	NULL	NULL	NULL

Display food items and their prices

Input:

```
202 •    SELECT Food_ID, Food_Name, Price
203      FROM Food_Item;
```

Output:



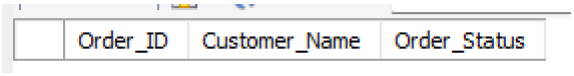
	Food_ID	Food_Name	Price
*	NULL	NULL	NULL

Verify foreign key relationships

Input:

```
205 •   SELECT
206         o.Order_ID,
207         c.Name AS Customer_Name,
208         o.Order_Status
209     FROM Orders o
210     JOIN Customer c
211         ON o.Customer_ID = c.Customer_ID;
```

Output:



	Order_ID	Customer_Name	Order_Status

These queries help verify stored records and demonstrate that related information can be retrieved through table relationships.

8. Result

Primary keys, foreign keys, UNIQUE, CHECK, and DEFAULT constraints were implemented or verified in the Online Food Delivery System database. SQL test statements were prepared to demonstrate how these constraints prevent duplicate values, invalid references, negative prices, and invalid quantities while assigning default values.

9. Conclusion

Data integrity constraints are essential for maintaining accurate and consistent database information. Primary keys identify records uniquely, foreign keys maintain relationships, UNIQUE constraints prevent duplicate values, CHECK constraints validate data, and DEFAULT constraints provide predefined values. These constraints improve the reliability of the Online Food Delivery System.